

Energy consumption estimates

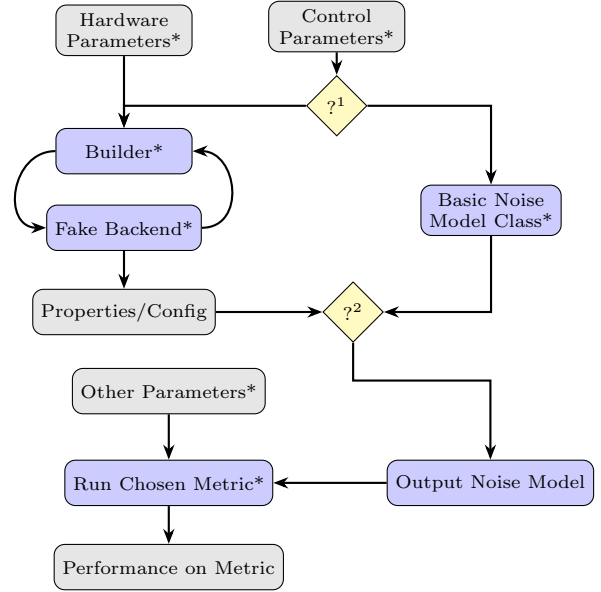
(Dated: February 11, 2026)

I. INTRODUCTION

II. SIMULATION RESULTS

To complement this mathematical approach, we also computed simulations tracking performance on quantum algorithms as we varied temperature as the single control parameter. We computed simulations of both:

- Full quantum circuits, tracking full density matrix representations, allowing us to employ full Qiskit noise models
- Circuits using only Clifford gates, tracking qubit state using stabiliser states and only employing simpler Pauli noise models. This allowed us to simulate much larger numbers of qubits.



?¹: Choose between an IBM Fake Backend or a basic noise model class
 ?²: A Pauli noise model is generated if the chosen metric is a Clifford metric. If a full noise model is generated, Pauli twirling can optionally be turned off.

FIG. 1. Dataflow for simulated performance on a quantum algorithm, from control parameters to noise model to performance. All nodes noted with a * are modular and customisable.

B. Case Studies

1. Case Study 1: *Quantum Volume on Legacy Backends*

A. Design Architecture

Something to be emphasised is the total modularity of the system we designed to do this. All parameters can be altered and all models can be swapped in and out, providing complete flexibility over how the performance value is computed.

Here is an overview of the system architecture:

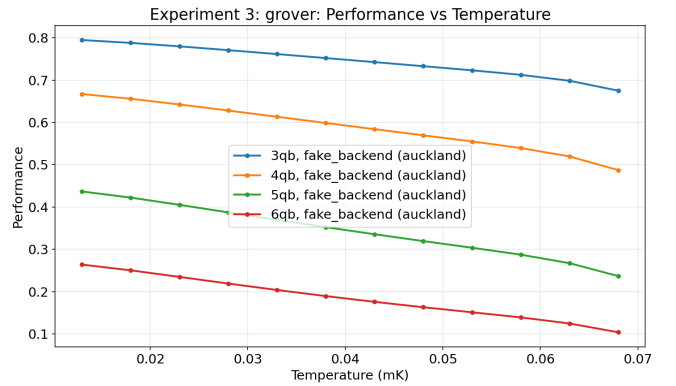


FIG. 2. Performance on Grover's algorithm using Auckland Fake Backend

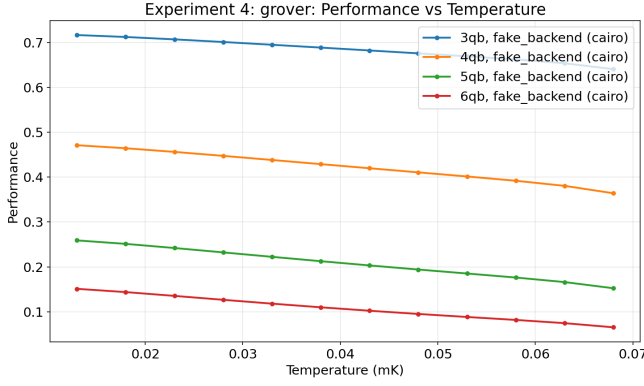


FIG. 3. Performance on Grover’s algorithm using CairoV2 Fake Backend

These are both V2, Falcon R5 backends released around late 2021. You can see a stable drop in performance with increased qubit counts.

C. Preface of Result Validity

The validity of these results hinges primarily upon the correctness of the code used to generate them. Thus, we have generated an extensive suite of unit tests to ensure the framework is operating as expected.

Moreover, the validity and applicability of these results further hinges upon the use of correct and properly catalogued assumptions used to generate them. We employ the following to mitigate the risks of compromising these qualities:

- The use of a variety of different methods of producing results, including a variety of different metrics and methods of generating noise models. These methods vary in complexity and specificity, avoiding how many assumptions are universally relied upon.
- Comprehensively cataloguing the assumptions used at every stage of generating the results, including how they change for each method of producing results.

To do this, we treat each set of results as a case study. Because there is so much overlap between the assumptions of each case study, we include a set of all set of assumptions in the appendix, and specify which ones apply for each.

There are a few assumptions that due to the inherent limitations of the simulation architecture universally apply, and are worth pointing out separately:

- All noise models are inherently gate-local (and thus also qubit-local)
- All noise models are state independent and unchanging over time throughout the whole metric simulation, and every shot and trial of the metric

- All noise models are Markovian

For all further assumptions, reference specific assumption sets on a per case study basis. The theory behind some of the assumptions is listed in the third subsection.

D. Case Studies

1. Case Study 1: Quantum Volume, Simple Arrhenius Noise Model

E. Assumptions

1. Hardware Parameters

I make the following modelling assumptions for the IBM processors:

All data mentioned below is IBM calibration data pulled directly from the latest github commit.

We use the following hardware parameters in the temperature dependent model:

- γ_0 is fit at runtime
 - It is calculated by minimising the squared error between the estimated and actual T1 values, keeping all other model parameters constant
 - To produce the estimated T1 values, we assume such data is generated at the default operational MXC temperature, which is taken as 0.013K
 - To retain per-qubit accuracy, we calculate and store the multiplicative error of each qubit’s T1 value at the operational temperature, and adjust all estimates for T1 values at other temperatures with this multiplicative factor
 - An improvement for future builder classes would be to store a value for γ_0 on a per qubit basis instead of averaging
- $\gamma_{\varphi\text{-base}}$ is fit at runtime
 - It is calculated by minimising the squared error between the estimated and actual T2 values, keeping all other model parameters constant
 - This is calculated after estimating γ_{a_0}
 - We assume such data is generated at operational MXC temperature, which is taken as 0.013K
 - As before, to retain per-qubit accuracy we calculate and store the multiplicative error of each qubit’s T2 value at the operational temperature and adjust all estimates for T2 values at other temperatures with this multiplicative factor

- An improvement for future builder classes would be to store a value for $\gamma_{\varphi-\text{base}}$ on a per qubit basis instead of averaging
- $\frac{\gamma_{MXC}}{\gamma_0}$ is estimated as 0.9
 - The maximum value from the experimental dataset reported in Ref. [1] is 0.85.
 - We can assume this is using standard lab equipment, making 0.9 a lower bound for professional IBM architecture
- T_{env} is fit at runtime
 - It is calculated by estimating a value for each qubit and taking the median
 - `prob_meas1_prep0` is extracted for each qubit, and used alongside previously estimated parameters and equations from [1] to estimate T_{env} , as described in Section II F
- Δ (superconducting energy gap) of 50,000
 - This is the known value for Aluminium, which we can assume the electrodes used in IBM transmon qubits are made of
- ω_{ge} is extracted from known configuration data for each backend individually for each qubit
- N_e , the number of effective channels in the Josephson Junction, is estimated at 1.29×10^5 . We calculate this value using:
 - An estimated normal resistance of the Josephson Junction of $5k\Omega$
 - Transparency of the junction in the subgap regime of 5×10^{-5}
- E_c , charging energy of the capacitors in the transmon qubits, of $0.95\mu\text{eV}$
- E_j , Josephson energy of the transmon qubits, of $62.1\mu\text{eV}$

When calculating gate errors, the difference between the known gate error and calculated gate error at the operational temperature is stored and assumed to be representative of the non-thermal error, and thus independent of temperature. It is then added to future calculated gate errors

F. Theory

T_{env} is calculated in the Default Builder using the following pathway:

- First the probability of measuring $|1\rangle$ after preparing $|0\rangle$ is extracted for each qubit
- The residual excited state population, p_e , as estimated as equal to this value.
- The number of effective thermal photons is estimated using from the Bose-Einstein distribution and assumption of normalisation ($p_e + p_g = 1$):

$$n_{\text{eff}} = \frac{p_e}{1 - 2 * p_e}$$

- Following the two-bath model of Ref. [1], n_{eff} is given by [Eq. (14)]:

$$n_{\text{eff}} = \frac{\gamma_{MXC}}{\gamma_1^0} n(\omega_{ge}, T_{MXC}) + \frac{\gamma_{\text{env}}}{\gamma_1^0} n(\omega_{ge}, T_{\text{env}})$$

- Now, taking $n(\omega_{ge}, T_{MXC})$ to be negligible and rearranging for T_{env} we get:

$$T_{\text{env}} = \frac{\hbar\omega_{ge}}{k_B \ln\left(1 + \frac{\gamma_{\text{env}}}{\gamma_1 n_{\text{eff}}}\right)}$$

- We can then use qubit specific values of ω_{ge} , and the γ_1 term cancels out

Note that this is an upper bound, and thus safely overestimates noise.

[1] D. S. Lvov, S. A. Lemziakov, E. Ankerhold, J. T. Peltonen, and J. P. Pekola, Physical Review Applied **23**, 054079 (2025).